Privacy-Preserving Detection of Cognitive Fatigue from Keystroke and Mouse Dynamics

1. Introduction

Cognitive fatigue can affect attention, information processing, and task performance during prolonged computer-based work. Conventional approaches to monitoring fatigue often use self-report measures or physiological signals, but these methods may be difficult to deploy continuously in everyday desktop environments. Keystroke and mouse dynamics offer a non-intrusive alternative by describing how a user interacts with standard input devices, rather than requiring additional sensors. However, these interactions are also behavioral biometrics and therefore raise privacy concerns. This review examines evidence for keyboard-based fatigue assessment, the potential role of mouse and multimodal interaction signals, and privacy-preserving approaches that could support a practical fatigue-detection system.

2. Keystroke Dynamics and Fatigue Detection

Existing evidence supports the hypothesis that typing patterns change with fatigue, although results depend strongly on the task and study setting. Ulinskas et al. [1] used keyboard press-and-release timing to classify levels of daytime fatigue and reported an average accuracy of 98.11% for a three-class formulation. This finding demonstrates potential under their experimental conditions; it should not be assumed to generalize to unconstrained everyday typing. In a natural office setting, de Jong et al. [2] monitored employees over six weeks and reported changes in typing speed and error-correction strategies across work periods and times of day. Their study highlights both the promise of unobtrusive measurement and the importance of context-dependent variation.

Acien et al. [3] adapted a neural keystroke model originally developed for biometric authentication to the mental-fatigue task. Across three keystroke databases, their fatigue-versus-rested analyses produced areas under the ROC curve of approximately 0.722–0.800. More recently, Bennett et al. [4] reported that multifractal properties of keystroke timing decreased after a mentally fatiguing task and were related to self-reported fatigue. Fang et al. [5] investigated smartphone typing patterns in relation to time awake, adding evidence from a different device context. Together, these studies indicate that useful information may exist in timing, irregularity, pauses, and error-related features, but they also show that fatigue prediction requires careful definition of the label and recording context.

3. Mouse Dynamics and Multimodal Behavioral Signals

Keyboard-only models cannot capture periods dominated by pointing, clicking, or browsing. Mouse-derived features such as movement speed, acceleration, pauses, click frequency, and

trajectory characteristics may therefore offer complementary information, although each proposed indicator requires validation against fatigue rather than unrelated changes in task demands. Sahana and Ananda Babu [6] reviewed techniques for mental fatigue detection using both keystroke and mouse dynamics and discussed traditional machine-learning classifiers and practical limitations of available datasets. This review establishes that the general combination of keyboard and mouse inputs is already present in the literature; merely combining the two modalities is not sufficient to establish novelty.

Research on behavioral authentication also informs multimodal feature integration, with an important limitation: identifying a user and detecting that user's fatigue are different tasks. Li et al. [7] developed a multimodal behavioral-biometric framework for continuous authentication combining mouse behavior and other interaction signals. While their work illustrates approaches to feature fusion, authentication performance cannot be interpreted as fatigue-detection accuracy. A fatigue-focused evaluation must therefore compare keystroke-only, mouse-only, and fused features using independently defined fatigue targets.

4. Privacy-Preserving Learning and Behavioral Biometrics

Keystroke timing and mouse movements can reveal distinctive personal behavior even when the actual typed text is excluded. Privacy, therefore, needs a concrete threat model rather than a general claim that the system is anonymous. Baig and Eskeland [8] proposed a privacy-preserving keystroke authentication protocol based on homomorphic cryptographic operations, demonstrating that behavioral features can be processed without directly exposing their plaintext values to an authentication server. However, the protocol addresses authentication rather than fatigue detection and would introduce implementation and computational considerations.

Federated learning is another relevant design option. McMahan et al. [9] demonstrated model training from decentralized data by aggregating locally computed updates, and Yang et al. [10] applied a federated approach to cross-device keystroke authentication while keeping typing-rhythm information locally. Crucially, federated learning alone does not mathematically guarantee that model updates reveal no information. Differentially private model training, as investigated by Abadi et al. [11], offers a formal framework for limiting information leakage, subject to privacy-utility trade-offs. For a first prototype, local feature computation, avoidance of text-content logging, strict retention limits, and on-device inference can be practical measures to minimize. Stronger claims of formal privacy would require a specified adversary, leakage evaluation, and appropriate privacy technology.

5. Research Gap and Proposed Direction

The reviewed literature supports fatigue-sensitive typing signals [1]–[5], existing work on keyboard–mouse fatigue modeling [6], and techniques for privacy-sensitive behavioral learning

[8]–[11]. Nevertheless, these strands do not by themselves establish a validated, privacy-preserving cognitive-fatigue detector that is reliable across users and ordinary desktop tasks. The proposed research gap is therefore the need to evaluate the trade-off between fatigue-detection effectiveness, privacy exposure, and practical deployment when keystroke and mouse features are combined. This is a provisional gap statement and must be refined after full-text screening of additional relevant studies.

A feasible research design would compare (i) keystroke-only, (ii) mouse-only, and (iii) fused-feature baselines under the same data collection and participant-independent evaluation protocol. Fatigue labels should be supported by a validated self-report instrument and, where feasible, an objective performance task; time spent typing or working should not be treated as a definitive fatigue label. The team should report macro-F1, fatigue-class recall, ROC-AUC or precision–recall metrics where appropriate, and results on held-out participants or sessions. Privacy should be operationalized through explicit data minimization and storage rules, with any stronger protection, such as federated learning or differential privacy, evaluated against accuracy and resource costs. Consent, collection boundaries, and avoidance of password or sensitive-field logging must be planned before data acquisition.

6. Conclusion

Prior studies show that ordinary input-device behavior can provide signals associated with fatigue, while behavioral-biometrics research demonstrates both opportunities for multimodal modeling and reasons for caution about privacy. The contribution of the proposed project should not be simply that fatigue can be inferred from typing or mouse movement. Instead, it should test whether a clearly specified privacy-conscious design can preserve useful fatigue-prediction performance under realistic cross-user and cross-task evaluation. The final research claim must be based on the team's own results rather than assumed from the literature.

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